



Chemistry: CYL1010

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Transition Metal Chemistry [5 Lectures]: Colors in Complexes, Organometallics and Catalysis, Bioinorganic Chemistry

Organic Chemistry [8 Lectures]: Name Reactions, Stereochemistry and their Applications, Acid-Base Chemistry, Use of Buffers, Organic Chemistry examples such as polymers and biopolymers, drugs and photoactive molecules in everyday life and advanced technologies

Acid

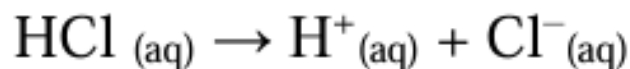
- Sour taste
- Reacts with carbonates to make CO_2
- Reacts with metals to produce H_2

Base

- Bitter taste
- Reacts with fats to make soaps
- Do not react with metals

Arrhenius Theory

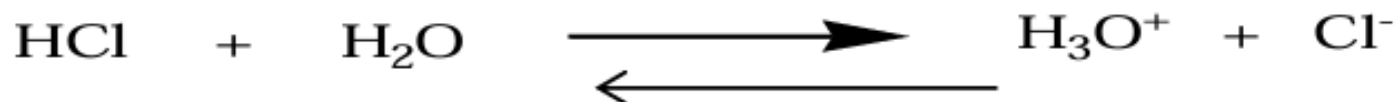
According to Arrhenius Concept, Acid are substance which are capable of providing Hydrogen ions (H^+ , proton) when dissolved in water and bases are substances which are capable of providing hydroxide ions (OH^- , hydroxyl ions) in aqueous solution.



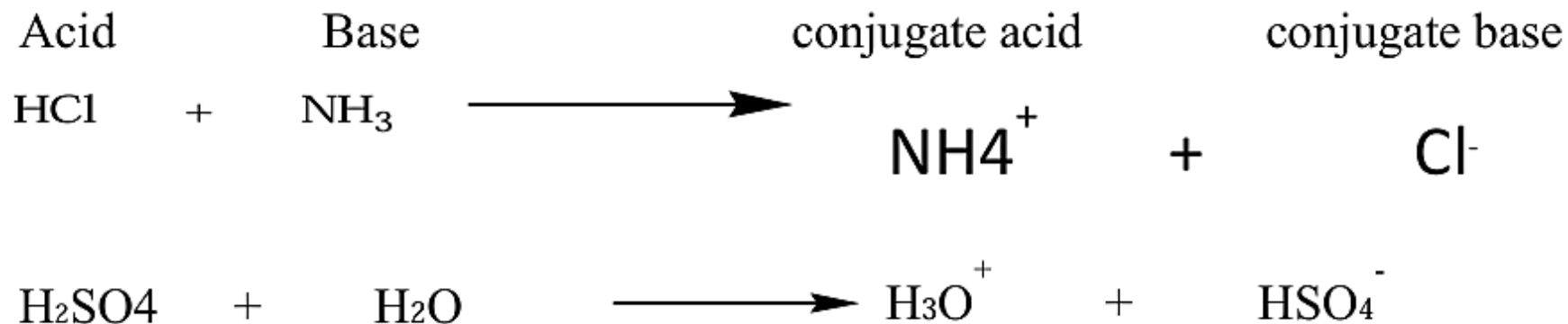
The Arrhenius theory is applicable only in aqueous solution

Bronsted -Lowry concept

An acid is any substance (molecular or ionic) that can donate a proton to any other substance (molecular or ionic) and a base is any substance that can accept a proton from any other substance.



For the reverse reaction Cl^- behaves as a Brønsted base and H_3O^+ behaves as a Brønsted acid. The Cl^- is called the conjugate base of HCl . Brønsted acids and bases always exist as conjugate acid-base pairs. Their formulas differ by only one proton.



- Every Arrhenius Acid is Bronsted Acid.

- Every Arrhenius Base is not Bronsted Base e.g., NaOH is Arrhenius base because it gives OH⁻ ion in aqueous solution but not a Bronsted base because it cannot accept proton.

Limitations of Bronsted Lowry Concept:

- The protonic definition cannot be used to explain the reactions occurring in non-protonic solvents such as COCl_2 , SO_2 , N_2O_4 , etc.
- Substances like BF_3 , AlCl_3 etc, do not have any hydrogen and hence cannot give a proton but are known to behave as acids

Lewis Acid Base concept

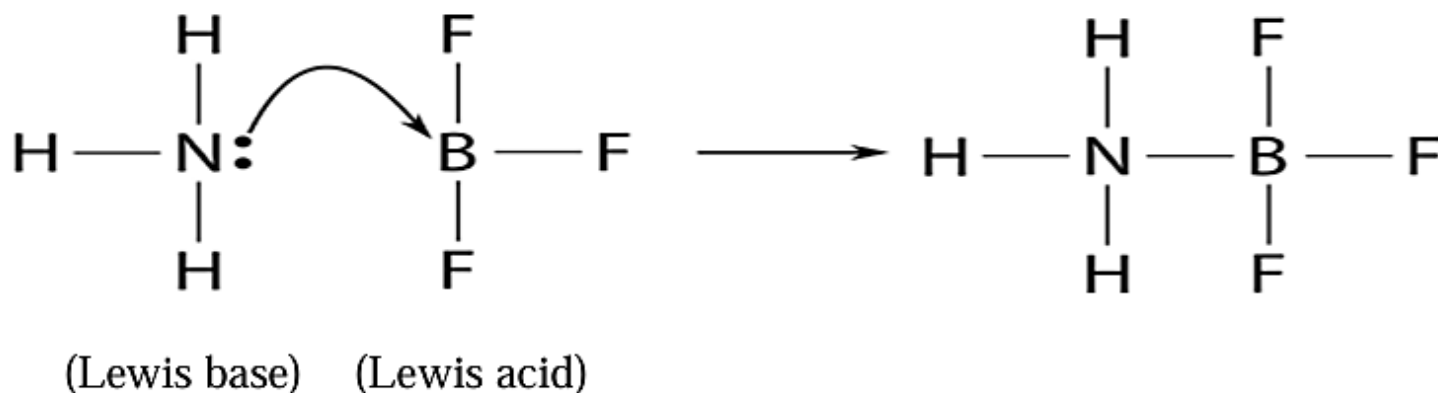
A base is an electron pair donor and acid is an electron pair acceptor.

Lewis Acids:

- Lewis acids accept an electron pair. Lewis Acids are Electrophilic meaning that they are electron attracting.
- Various species can act as Lewis acids. All cations are Lewis acids since they are able to accept electrons. (e.g., Cu^{2+} , Fe^{2+} , Fe^{3+})
- **Lewis acids-** H^+ , NH_4^+ , Na^+ , K^+ , Cu^{2+} , Al^{3+} , etc.

Lewis Bases

- Lewis Bases donate an electron pair. Lewis Bases are [Nucleophilic](#) meaning that they “attack” a positive charge with their lone pair. An atom, ion, or molecule with a lone-pair of electrons can thus be a Lewis base.
- **Lewis base-** NH_3 , H_2O , OH^- , Cl^- , CN^- , S^{2-} , etc.
- The reaction of boron trifluoride with ammonia is an example.



Boron trifluoride **accepts** the electron pair, so it is a **Lewis acid**. Ammonia makes available (donate) the electron pair, so it is the Lewis base.

Buffer

Buffers: Buffers are defined as a compound or a mixture of compounds that resists the pH upon the addition of small quantities of acid or alkali. Buffer have definite pH value. The pH will not change after keeping it for a long period of time. The pH value altered negligibly by the addition of small quantities of acid or base.

Buffer action: The resistance to a change in pH is known as buffer action. So buffers can be added to show buffer action.

Buffer capacity: The amount of acid/base required to produce a unit change in pH in a solution is called buffer capacity.

Types of buffers:

Generally buffers are of two types:

1. Acidic buffers
2. Basic buffers

There are some other buffer system:

3. Two salts acts as acid-base pair. Ex- Potassium hydrogen phosphate and potassium dihydrogen phosphate.
4. Amphoteric electrolyte. Ex- Solution of glycine.

1. Acidic Buffers:

An acidic buffer is a combination of weak acid and its salt with a strong base.

i.e. Weak acid & salt with strong base (conjugate base).

☐ EXAMPLES:

- $\text{CH}_3\text{COOH} / \text{CH}_3\text{COONa}$
- $\text{H}_2\text{CO}_3 / \text{NaHCO}_3$
- $\text{H}_3\text{PO}_4 / \text{NaH}_2\text{PO}_4$
- $\text{HCOOH} / \text{HCOONa}$

2. Basic Buffers:

A basic buffer is a combination of weak base and its salt with a strong acid.

i.e. Weak base & salt with strong acid (conjugate acid).

☐ EXAMPLES:

- $\text{NH}_4\text{OH} / \text{NH}_4\text{Cl}$
- $\text{NH}_3 / \text{NH}_4\text{Cl}$
- $\text{NH}_3 / (\text{NH}_4)_2\text{CO}_3$

Mechanism of Buffer action:

The resistance of a buffer solution to a change in pH is known as buffer action.

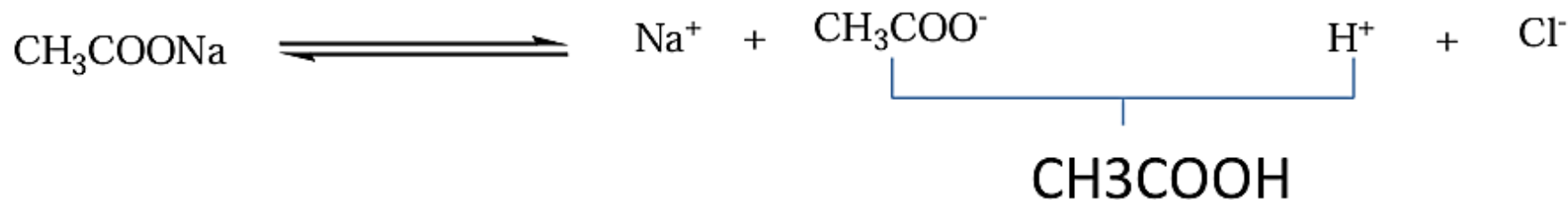
In a buffer solution, the components interact with each other and produce a dynamic equilibrium. When a small quantity of acid or base is added, the dynamic equilibrium shifts and nullifies the effect of the addition.

Mechanism of Action of acidic buffers:

- ❑ Consider a buffer system of CH_3COOH (Weak electrolyte) and CH_3COONa (Strong electrolyte). There will be a large concentration of Na^+ ions, CH_3COO^- ions, and undissociated CH_3COOH molecules.

When an acid is added:

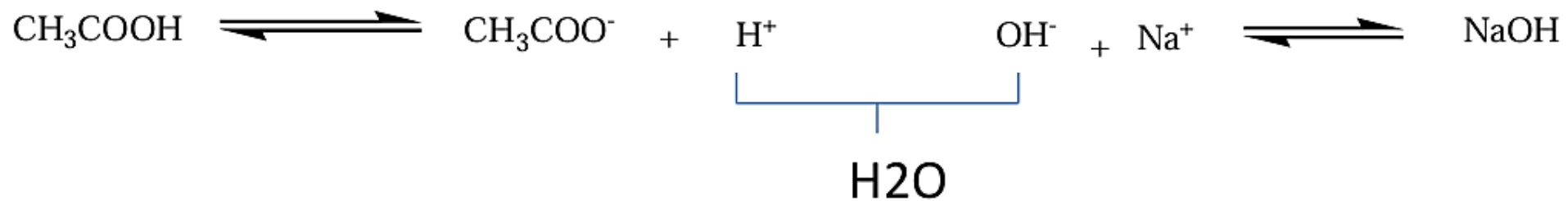
- If a strong acid (HCl) is added in $\text{CH}_3\text{COOH} / \text{CH}_3\text{COONa}$ buffer, the changes that will occur may be represented as:



- ❑ The hydrogen ions yielded by the HCl are quickly removed as unionized acetic acid, and the hydrogen ion concentration is therefore only slightly affected (because acetic acid produced is very weak as compared to HCl added).

When a base is added:

- If a strong base (NaOH) is added in CH_3COOH / CH_3COONa buffer, the changes that will occur may be represented as:



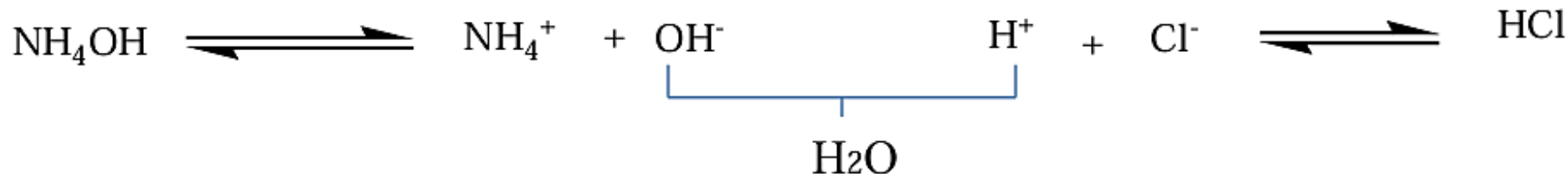
- ❑ The hydroxyl ions yielded by the NaOH are therefore removed as water. The supply of hydrogen ions needed for this purpose being constantly provided by the dissociation of acetic acid.

Mechanism of Action of basic buffers:

- ❑ Consider a buffer system of NH_4OH (Weak electrolyte) and NH_4Cl (Strong electrolyte). There will be a large concentration of NH_4^+ ions, Cl^- ions, and undissociated NH_4OH molecules.

When an acid is added:

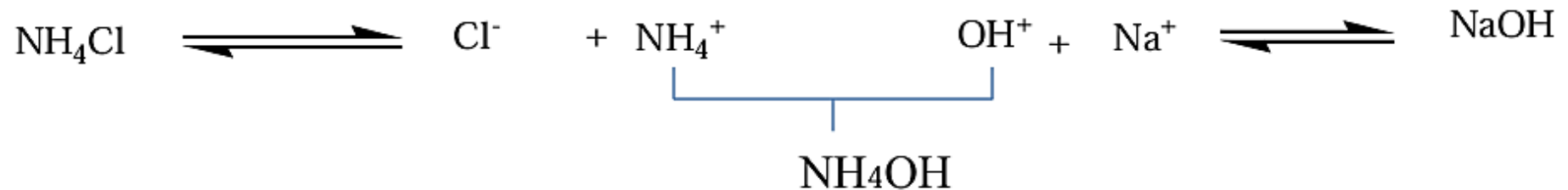
- If a strong acid (HCl) is added in $\text{NH}_4\text{OH} / \text{NH}_4\text{Cl}$ buffer, the changes that will occur may be represented as:



- ❑ The hydrogen ions yielded by the HCl are therefore removed as water. The supply of OH^- ions needed for this is constantly provided by the ammonium hydroxide.

When a base is added:

- If a strong base (NaOH) is added in NH_4OH / NH_4Cl buffer, the changes that will occur may be represented as:



- ❑ The hydroxyl ions yielded by the NaOH are therefore quickly removed as unionized ammonium hydroxide and the pH of solution is only slightly affected.

Buffer equation-Henderson-Hasselbalch equation:

The buffer equation is also known as Henderson-Hasselbalch equation, with the help of this equation it is possible to calculate the pH of a buffer solution of known concentration or to make buffer solution of known pH.

Two separate equations are obtained for each type of buffer, acidic and basic

pH of acidic buffer: The hydrogen ion concentration obtained from the dissociation of weak acid HA is given by equation,



$$K_a = \frac{[\text{H}^+][\text{A}^-]}{[\text{HA}]}$$

K_a = equilibrium constant

$$[\text{H}^+] = K_a \frac{[\text{HA}]}{[\text{A}^-]}$$

Taking logarithms of both sides of the equation & multiplying throughout by -1 gives

$$-\log[\text{H}^+] = -\log K_a - \log \frac{[\text{HA}]}{[\text{A}^-]}$$

$$pH = pKa + \log \frac{[A^-]}{[HA]}$$

$$pH = pKa + \log \frac{[\text{conjugate base}]}{[\text{acid}]}$$

pH of an alkaline buffer: The ionization of a weak base BOH is given by,



$$Kb = \frac{[B^+][OH^-]}{[BOH]}$$

$$[OH^-] = Kb \frac{[BOH]}{[B^+]}$$

$$\log[OH^-] = \log Kb + \log \frac{[BOH]}{[B^+]}$$

$$-\log[OH^-] = -\log Kb + \log \frac{[B^+]}{[BOH]}$$

$$pOH = pKb + \log \frac{[\text{conjugate acid}]}{[\text{base}]}$$

Now,

$$\begin{aligned} pH &= 14 - pOH \\ &= 14 - \left(pKa + \log \frac{[\text{conjugate acid}]}{[\text{base}]} \right) \end{aligned}$$

Buffer capacity:

Buffer capacity may also be defined as “*The maximum amount of either strong acid or strong base that can be added before a significant change in the pH will occur*”.

The maximum amount of strong acid that can be added is equal to the amount of conjugate base present in the buffer whereas the maximum amount of base that can be added is equal to the amount of weak acid present in the buffer.

- Buffer capacity is depend on the factors:

1. The concentration of the acid or base component of the buffer (Direct relation)
2. The pH of the buffer

➤ Buffer can act best at $\text{pH} = \text{pK}_a$ and buffering range is $\text{pH} = \text{pK}_a \pm 1$

Or It may be defined as the moles of strong acid or strong base required to change the pH of 1000 ml of buffer solution by one unit.

- The magnitude of the resistance of a buffer to pH changes is referred to as the buffer capacity, β .

$$\beta = \frac{\Delta B}{\Delta \text{pH}}$$

Where, ΔB is the small increment in gram equivalents (g Eq)/liter of strong base added to the buffer solution and ΔpH : change in a pH

Standard Buffer Solutions:

The standard buffer solutions of pH ranging from 1.2-10 are possible to prepare by appropriate combinations of 0.2N HCl or 0.2N NaOH or 0.2M solution of potassium hydrogen phthalate, potassium dihydrogen phosphate, boric acid, potassium chloride.

Standard buffers with pH range:

Buffer	pH
Hydrochloric acid buffer	1.2-2.2
Acid Phthalate buffer	2.2-4.0
Neutralised phthalate buffer	4.2-5.8
Phosphate buffer	5.8-8.0
Alkaline Borate buffer	8-10

Preparation of Buffer Solutions:

Weak acid with $pK_a = \text{desired pH}$ should be selected



Ratio of salt and acid needed should be calculated using buffer equation



Individual concentration of buffer salt and acid determined.



Ingredients are dissolved in carbon dioxide free water



Buffer capacity of 0.01-0.1 is adequate



Concentration of 0.05-0.5 M is sufficient



Allowed to establish equilibrium



pH verified.

Application of Buffers

Dosage Form	Application	Buffer used
Solids (Tablets, Capsules, powder)	Control pH for release. Reduce gastric irritation	Citrate buffer, Phosphate buffer
Semisolids (Creams, Ointments)	Stability	Citric acid and sodium citrate
Parenteral Products	pH maintenance (pH high: tissue, necrosis pH low: pain)	Citrates, glutamate, acetate, phthalate
Ophthalmic products	Drug solubility and stability	Borates, carbonates, phosphates

1. Solubility enhancement: The pH of the pharmaceutical formulations are adjusted to an optimum value so that the drug remains solubilised throughout its shelf-life and not precipitated out.

Eg. Sodium salicylate (Asprin) precipitates as salicylic acid in acidic environment.

2. Increasing stability: To prevent hydrolysis and for maximum stability, the pH of the medium should be adjusted suitably. **Eg.** Vitamins

3. Improving purity: The purity of proteins can be identified from its solubility at their isoelectric point as they are least soluble at this point. The isoelectric pH can be maintained using suitable buffers. **Eg.** Insulin precipitates from aqueous solution at pH 5.0-6.0.

4. Optimising biological activity: Enzymes have maximum activity at definite pH values. Hence buffer of desired pH is added to the preparation.

5. Comforting the body: The pH of the formulations that are administered to different tissues of the body should be optimum to avoid irritation (eyes), haemolysis (blood) or burning sensation (abraded surface). The pH of the preparation must be adjusted with suitable amount of buffers to match with the pH of the physiological fluid.